

A Hybrid Multi-Paradigm Weighting Framework for Composite Sustainability Indices: Integrating Structural Equation Modeling, Analytic Hierarchy Process, and AI-Based Optimization

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Abstract

Composite sustainability indices are widely used to synthesize multidimensional performance into actionable decision-support tools. However, the methodological legitimacy of such indices remains critically dependent on the weighting scheme adopted. Existing approaches typically rely on either empirical techniques (e.g., Structural Equation Modeling), normative multi-criteria decision-making methods (e.g., Analytic Hierarchy Process), or data-driven optimization frameworks rooted in artificial intelligence. Each paradigm embodies distinct epistemological assumptions, yet no single approach fully reconciles theoretical validity, stakeholder interpretability, and adaptive robustness.

This paper proposes a hybrid multi-paradigm weighting framework that integrates (i) empirically derived latent-structure weights via Structural Equation Modeling, (ii) expert-driven normative weights using the Analytic Hierarchy Process, and (iii) adaptive weights optimized through artificial intelligence-based learning algorithms. The three weighting vectors are aggregated through a meta-weight architecture, allowing strategic calibration of epistemic contributions. A formal mathematical formulation of the hybrid index is provided, followed by a sensitivity analysis of meta-weights and a Monte Carlo simulation to assess robustness under uncertainty.

Furthermore, a Gap-Based Sustainability Assessment Framework is introduced to operationalize the hybrid index in policy diagnostics, enabling identification of structural performance gaps across sustainability dimensions. The results demonstrate that the hybrid approach improves stability, interpretability, and responsiveness compared to single-paradigm weighting schemes.

The proposed framework contributes theoretically by reconciling competing weighting philosophies within a unified architecture and practically by offering policymakers a robust, adaptive, and strategically tunable sustainability measurement tool.

Keywords: Composite Sustainability Index; Hybrid Weighting Framework; Structural Equation Modeling; Analytic Hierarchy Process; Artificial Intelligence Optimization; Monte Carlo Sensitivity Analysis.

tional policies, but must also incorporate how these initiatives are perceived and evaluated by stakeholders. In this perspective, sustainability is not only an operational outcome but also a perceptual and relational construct shaped by stakeholder experience and interpretation.

The qualitative results highlighted a systematic divergence between stakeholder perceptions of existing sustainability initiatives and their expectations regarding institutional sustainability performance. While several operational actions were identified across economic, social, and environmental domains, stakeholders frequently emphasized limitations related to their visibility, institutionalization, and strate

1 Introduction

Sustainability assessment has become a central concern in higher education governance. Universities are increasingly expected to demonstrate accountability not only in terms of academic performance, but also regarding their economic efficiency, social responsibility, and environmental impact. As a result, composite sustainability indices have emerged as widely adopted tools for synthesizing multidimensional institutional performance into interpretable decision-support metrics.

Despite their growing popularity, the methodological legitimacy of composite sustainability indices remains contested. A central challenge concerns the determination of weights assigned to sustainability dimensions. Weighting schemes significantly influence index outcomes, institutional rankings, and policy interpretations. Traditional approaches typically rely on equal weighting, empirically derived statistical coefficients, or expert-based multi-criteria decision-making techniques. However, each of these approaches embodies distinct theoretical assumptions and epistemological orientations.

Empirical weighting methods, such as Structural Equation Modeling (SEM), derive weights from latent structural relationships among sustainability constructs. While statistically grounded, such methods may insufficiently reflect stakeholder normative priorities. In contrast, normative approaches, such as the Analytic Hierarchy Process (AHP), incorporate stakeholder judgments through structured pairwise comparisons, enhancing legitimacy but potentially introducing subjectivity. More recently, Artificial Intelligence (AI)-based optimization mechanisms have enabled adaptive and context-sensitive weighting schemes, although concerns regarding interpretability and transparency persist.

Beyond weighting challenges, conventional sustainability indices often conceptualize sustainability as a performance-based outcome measured through operational indicators. Such perspectives may overlook the relational dimension of sustainability, particularly within institutional contexts where stakeholder expectations shape legitimacy and perceived effectiveness. Sustainability in universities cannot be fully captured by operational outputs alone; it must also account for the alignment between institutional initiatives and stakeholder priorities.

To address these limitations, this study proposes a Gap-Based Sustainability Assessment Framework for universities. Sustainability is reconceptualized as an alignment-based construct, defined by the degree of coherence between stakeholder expectations and perceived institutional performance across economic, social, and environmental dimensions. This perspective shifts the analytical focus from static performance measurement to dynamic alignment evaluation.

Building upon this conceptual foundation, the study introduces a Hybrid Multi-Paradigm Weighting Framework integrating three complementary approaches: (i) empirically derived weights through Structural Equation Modeling, (ii) normative stakeholder-based weights through the Analytic Hierarchy Process, and (iii) adaptive weights optimized via Artificial Intelligence techniques. These weighting vectors are combined through a meta-weight structure, allowing strategic calibration of empirical validity, normative legitimacy, and contextual adaptability.

To enhance methodological robustness, the proposed framework incorporates both deterministic sensitivity analysis and Monte Carlo simulation to evaluate the stability of the hybrid index under meta-weight uncertainty. This probabilistic robustness assessment transforms the weighting structure from a fixed assumption into an explicitly modeled governance parameter.

The contribution of this research is threefold. First, it advances sustainability theory by

introducing an alignment-based conceptualization of institutional sustainability. Second, it reconciles empirical, normative, and adaptive weighting paradigms within a unified mathematical architecture. Third, it strengthens composite index methodology through explicit sensitivity and stochastic robustness modeling.

The remainder of this paper is structured as follows. Section 2 presents the conceptual foundation of the gap-based sustainability model. Section 3 describes the methodological design and measurement model specification. Section 4 formulates the University Sustainability Index. Section 5 introduces the hybrid weighting framework. Section 6 presents sensitivity and Monte Carlo robustness analysis. Section 7 discusses theoretical and governance implications, and Section 8 concludes.